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Benoit Caillaud (Inria)

Subject: Application for PhD Position – Structural Methods for Mixed Model/Data Digital Twin Engineering

Dear Benoit Caillaud,

I am applying for the PhD position on "Structural Methods for Mixed Model/Data Digital Twin Engineering" with great enthusiasm. With a strong engineering background in Artificial Intelligence, distributed computing, and complex system modeling, I am highly motivated to contribute to the Artemis framework and tackle the unique challenges presented by this project.

Bridging first-principles physical modeling (like Modelica) with data-driven streams requires rigorous structural methods. In my portfolio, I have engineered systems that combine analytical equations (Brayton Cycle) with Edge-TCN data inference. I am eager to formalize these hybrid structures, researching new structural methodologies that allow mixed model-data systems to be synthesized, verified, and safely executed.

I am particularly drawn to this position because it bridges rigorous research with practical industrial challenges within the EDT program. Please find attached my resume and academic transcripts, which detail my technical competencies and engineering background. I welcome the opportunity to discuss my application further.

Sincerely,

Ismail Aissaoui